



REQUIREMENTS SPECIFICATION

Favourite Books Online Store

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1. Introduction

Favourite Books is a bookstore located on Glenferrie Road. They have identified the popularity of online bookstores and wish to branch out into this area to increase their business reach.

This specification document will describe an online bookstore system that will support the operations of Favourite Books, including book browsing, purchasing, and management of orders and inventory.

The purpose of this document is to outline the requirements and quality attributes of the proposed online bookstore system for Favourite Books, which is intended to support the transition from a physical store to an online platform and reduce the limitations of the current manual processes. The functional requirements have been identified and organised into user tasks using the Tasks Description approach, focusing on what users need to perform within the system. An overview of the domain model is provided to show the relationships between key entities and actors. Quality attributes are included to ensure the system meets requirements such as security and usability. Finally, multiple possible solutions are considered to demonstrate how the system may be implemented.

2. Project Overview

2.1 Domain Vocabulary

Customer – Potential clients who browse the system for books.

Book – The product being sold.

Catalogue – A collection of books available for customers to view.

Cart – A temporary list where selected books are stored before checkout.

Order – Created when a customer confirms a purchase.

Payment – The process of completing payment for an order.

Invoice – A record showing the details and total cost of an order.

Shipment – The delivery of books to the customer after purchase.

2.2 Goals

Beyond just increasing the market reach of the store, the system also aims to centralise data regarding supply and sales statistics. To meet all these goals, the system must:

- Enable customers to browse the book catalogue online
- Support account creation and management for customers
- Provide shopping cart and order placement functionality
- Create invoices and receipts for customers
- Manage packing and shipment logistics
- Provide daily, weekly, monthly, and yearly sales statistics
- Allow the bookstore to update the catalogue

2.3 Assumptions

It can be assumed that customers have access to the internet and basic computer literacy skills, that the store has the necessary inventory and resources to support sales and delivery, and that many existing customers will be creating accounts to use the new online system.

As the system relies upon online transactions, it is assumed that payments will be handled through a secure external service. If payments are not managed properly, this may result in failed transactions or security risks, which could affect both customers and the business. For this reason, the orders will only be confirmed after payment has been successfully completed.

It can also be assumed that proper security measures are in place to protect customer data and transactions from unauthorised access or potential breaches, and access will be restricted so that only the owner/administrator has permission to update and manage the system, while staff are limited to handling packaging and order preparation.

2.4 Scope

The project cannot be expected to draw from a large budget, as it is being developed for a small store. The service must operate securely as to protect customer data and payment information. For the time being, discounts, loyalty programs, and rent-before-buying systems are not to be implemented, but may be considered in the future, and as such the system should be developed in a way to flexibly facilitate such an expansion of scope.

3. Domain Analysis

3.1 Pain Points

- Limited Accessibility
 - Customers are required to visit the physical store to browse and purchase books
 - This restricts convenience and reduces overall reach
- Inefficient Processes
 - Orders and stock are handled manually
 - As the business grows, this can lead to delays and *increase the chance of mistakes*
- Data Management Issues
 - Information is not stored in a central system
 - Difficulty in keeping records consistent and easily accessible
- Lack of Insights
 - No clear way to monitor sales or performance over time
- Operational Limitations
 - System access is limited
 - Difficulty for owners/staff to manage tasks at the same time

3.2 Domain Entities:

- Book
- Catalogue
- Order
- Order Item
- Customer
- Payment
- Invoice and Receipt
- Shipment
- Category
- Cart
- Inventory/Stock
- Sales Report

3.3 Actors

- Customer
- Administrator/Owner
- Employee
- Payment Service Provider(s)

3.4 Tasks List

Task 1: Create Customer Account

Task 2: Sign in to System

Task 3: Search and Browse Books

Task 4: View Books Details

Task 5: Add Books to Cart

Task 6: Modify Shopping Cart

Task 7: Review and Confirm Order

Task 8: Complete Payment Process

Task 9: Generate Invoice and Receipt

Task 10: Check Order Process

Task 11: Manage Book Listing and Classification

Task 12: Order Packaging and Delivery

Task 13: Analyse Sales Performance

4. Task Descriptions

Task 1: Create Customer Account

Task: 1	Register Customer Account
Purpose:	Register a new account
Trigger/Precondition	Customer does not have existing account
Frequency:	Once for each new customer registering an account
Critical:	Customer enters an existing account email or invalid details
Work Area:	Website
Sub-tasks:	
1. Customer inputs details	Registration form
2. Validate input	Check that required fields have been filled such as email, name, password etc
3. Verify input	Verify email
4. Create account	Store details securely in the system database
5. Confirm account creation	Prompt confirmation of successful registration
Variants:	
2a. Missing essential details	Highlight required field
3a. Existing customer account	Prompt account status (as existing customer)

Task 2: Sign in to System

Task: 2	Account Sign In
Purpose:	Access to an existing account
Trigger/Precondition	Customer has an existing account
Frequency:	Low, usually once per session
Critical:	Invalid username or password
Work Area:	
Sub-tasks:	
1. Enter username and password	Prompt user for login details
2. Validate credentials	Scan the database for matching credentials
3. Grant Access	Start session
4. Redirect to main page	System loads the user interface
Variants:	

Task 3: Search and Browse Books

Task: 3	Browse Book Catalogue
Purpose:	Allow customers to locate books within the catalogue
Trigger/Precondition	Customer accesses the website
Frequency:	High
Critical:	Handling large volumes of data efficiently
Work Area:	
Sub-tasks:	
1. View available books	System displays a list of books in the catalogue
2. Apply search or filter options	System provides search and filtering features
3. Display matching results	System shows results based on customer input
4. Select a book	System opens the selected book details
Variants:	
3a. No matching results	System displays a message indicating no results found

Task 4: View Book Details

Task: 4	View Book Details
Purpose:	Customer to view information
Trigger/Precondition	Customer clicks or selects a book
Frequency:	High
Critical:	Missing or incorrect book details
Work Area:	Book description page
Sub-tasks:	
1. Open the selected book	Load the book page
2. Show book details (title, price, author)	Information is displayed clearly
3. Show stock status	Display available stock
4. Choose next action (add to cart or return to main page)	Customer can add or exit the page
Variants:	
3a. Book is out of stock	Display a message to inform the user, and option to notify when available

Task 5: Add Books to Cart

Task: 5	Add Book to Cart
Purpose:	Allow the customer to add a book into the cart for buying
Trigger/Precondition	Customer is viewing a book
Frequency:	High
Critical:	Book not available
Work Area:	
Sub-tasks:	
1. Choose quantities	Customer enters quantity
2. Add book into cart	Put the book into cart
3. Update total prices	Update the cart total
4. Show confirmation	Prompt message confirming book added to card
Variants:	
1a. Book out of stock	Add book to cart action is disabled and shows a notice

Task 6: Modify Shopping Cart

Task: 6	Update Shopping Cart
Purpose:	Allow user to review and modify their cart before purchase
Trigger/Precondition	Shopping cart contains one or more items
Frequency:	Low, usually only a few times per session
Critical:	Incorrect total price calculation
Work Area:	
Sub-tasks:	
1. View cart items	Display all items currently in the cart
2. Change quantities	Update the quantity of selected items
3. Remove books	Remove the selected book from the cart
4. Recalculate total prices	Update the total price accordingly
Variants:	
1a. Shopping cart is empty	Prompt error message to notify user

Task 7: Review and Confirm Order

Task: 7	Checkout Order
Purpose:	Allow customers to confirm and place their order
Trigger/Precondition	Shopping cart contains one or more items
Frequency:	When customer proceeds to place an order
Critical:	Missing or incomplete required information
Work Area:	
Sub-tasks:	
Enter delivery details	Customer inputs the address form
Review order summary	System displays all selected books and total cost
Confirm order details	Customer verifies details and proceeds to checkout
Variants:	
Required fields missing	System prompts an error message and prevents checkout from continuing

Task 8: Complete Payment Process

Task: 8	Process Payment
Purpose:	Verify payment and complete purchase
Trigger/Precondition	Check out process is completed
Frequency:	Low, around once per session or less
Critical:	Payment transaction fails
Work Area:	
Sub-tasks:	
1. Enter payment details	Customer inputs payment information
2. Verify payment	Send the request to a third-party payment service and will confirm whether payment is successful or failed
3. Record transaction	System updates and stores payment details in the system
4. Update stock level	System updates the available quantity after successful payment
Variants:	
2a. Payment failure	System displays an error message for a failed payment and allows customer to retry with a different card/payment method

Task 9: Generate Invoice and Receipt

Task: 9	Generate Invoice and Receipt
Purpose:	Provide user with proof of purchase
Trigger/Precondition	Payment is successfully completed
Frequency:	After each successful payment
Critical:	Incomplete or incorrect transaction records
Work Area:	
Sub-tasks:	
1. Generate invoice	System creates an invoice record for the order
2. Generate receipt	System creates a receipt for the completed payment
3. Store records	System saves the transaction data in the database
4. Display and send receipt	Displays the receipt and sends a copy to customer's email
Variants:	

Task 10: Check Order Process

Task: 10	Track Order Status
Purpose:	Customers can check the delivery progress of their order
Trigger/Precondition	An existing order is available
Frequency:	Low
Critical:	Outdated or incorrect status
Work Area:	
Sub-tasks:	
1. View order history	System displays all user orders
2. Select an order	Customer chooses a specific order to view the status
3. View order status	System shows the current status and ETA (e.g. processing, shipped, delivered)
Variants:	

Task 11: Manage Book Listing and Classification

Task: 11	Manage Book Catalogue and Categories
Purpose:	Allows administrators to manage and update book catalogue and categories
Trigger/Precondition	Administrator has logged into the system
Frequency:	When administrators update or maintain the catalogue
Critical:	Inconsistency or incorrect information
Work Area:	
Sub-tasks:	
1. Add or edit book details	Admin inputs or updates book information or stock
2. Assign book category	System connects the book to the selected category
3. Monitor stock level	System displays stock quantity and highlights low stock items
4. Save changes	System stores and updates the changes
5. Display changes	System reflects changes
Variants:	

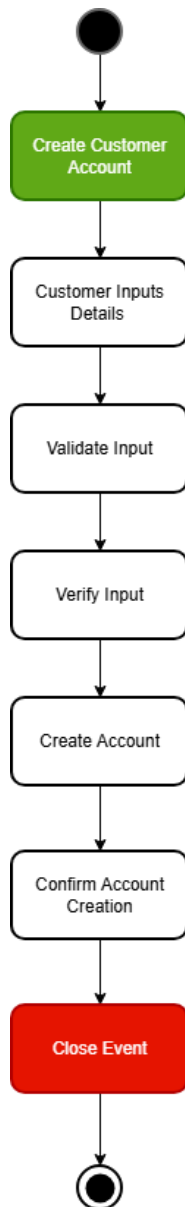
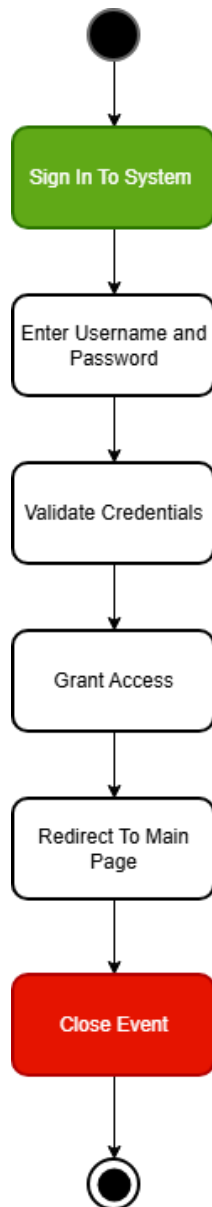
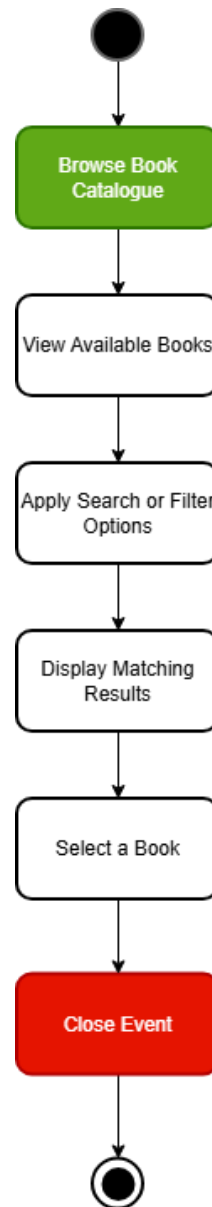
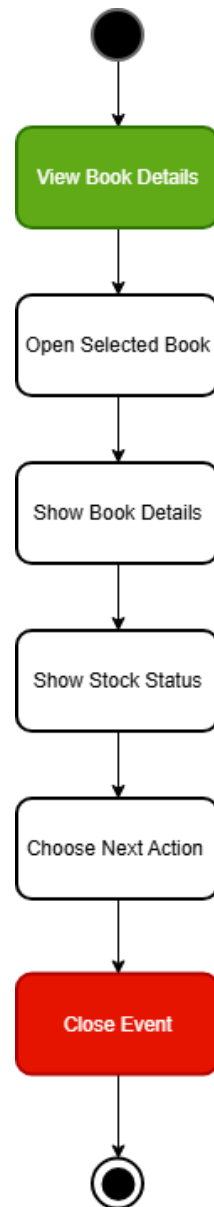
Task 12: Order Packaging and Delivery

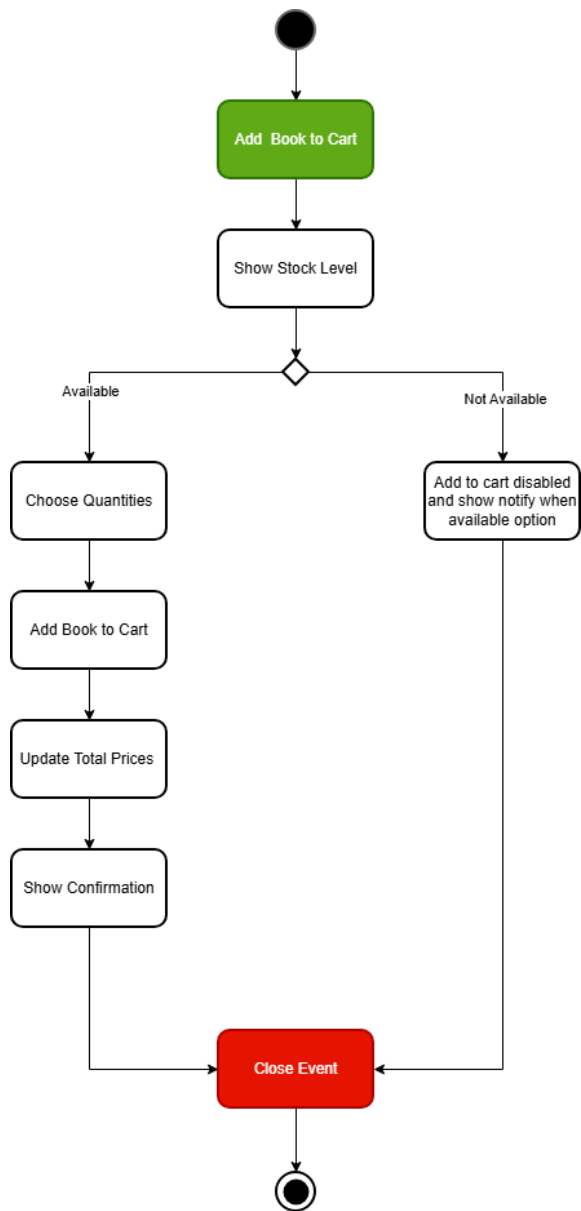
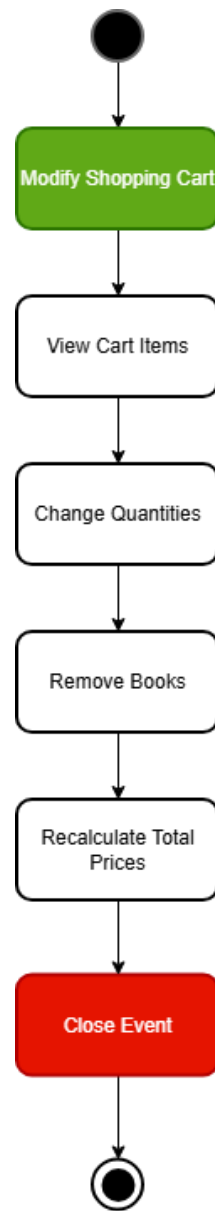
Task: 12	Process Packaging and Shipment
Purpose:	Ensure orders are prepared and delivered to customers
Trigger/Precondition	Order has been successfully paid
Frequency:	After an order is processed for delivery
Critical:	Incorrect delivery or shipment details
Work Area:	
Sub-tasks:	
1. View paid orders	System displays a list of successful purchase orders ready for processing
2. Package items	Staff prepare and pack the selected books
3. Create shipment record	System generates a shipment entry for the order
4. Update delivery status	System marks the order as shipped, provides tracking number and updates ETA
Variants:	

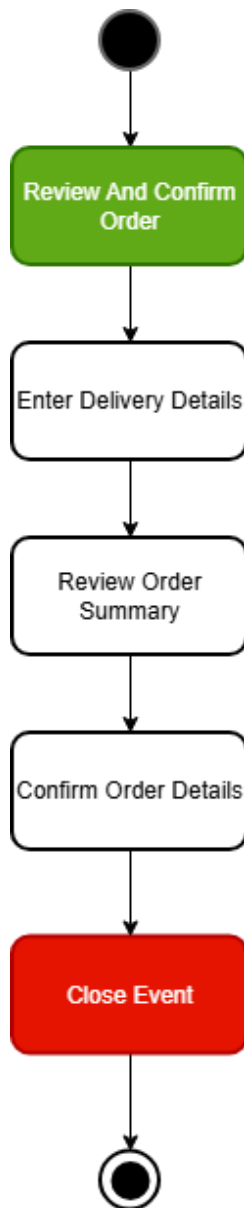
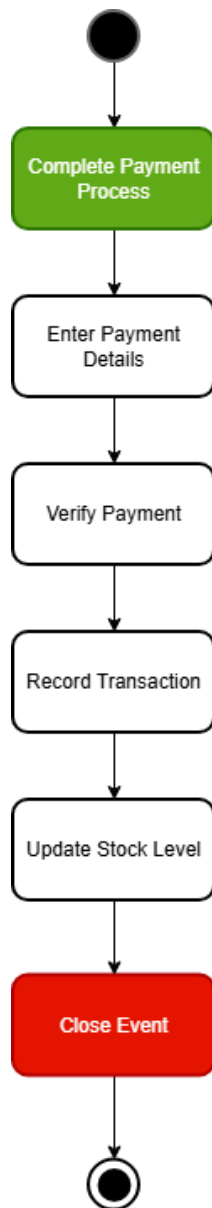
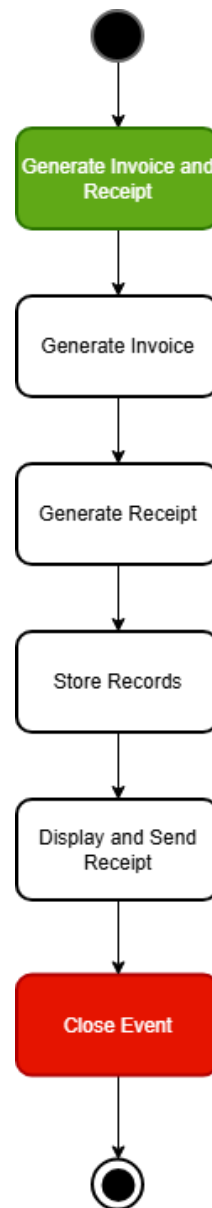
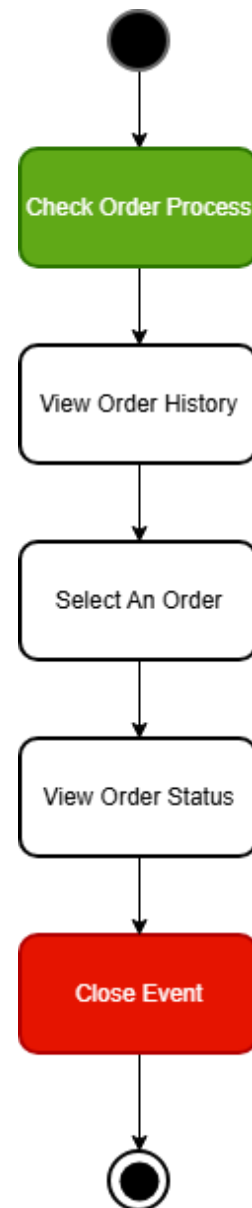
Task 13: Analyse Sales Performance

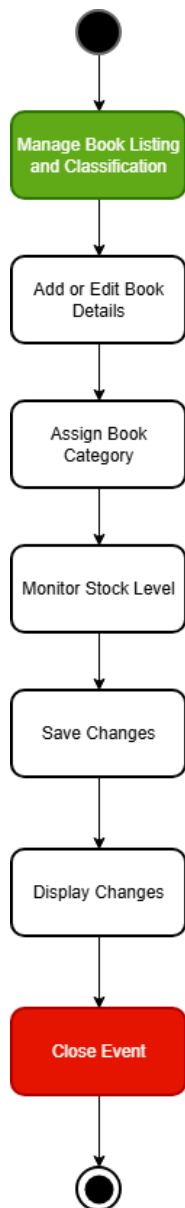
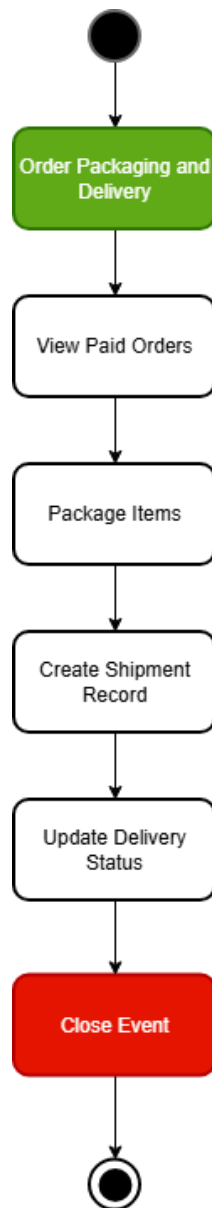
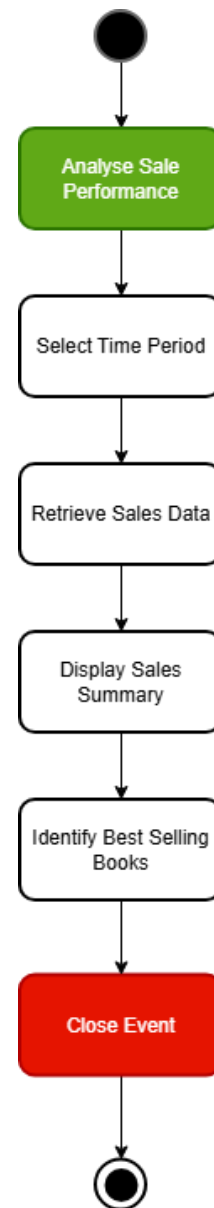
Task: 13	View Sales Statistics
Purpose:	Allow administrators to monitor sales performance
Trigger/Precondition	Sales data is available in the system
Frequency:	Daily when administrators review performance
Critical:	Incomplete data
Work Area:	
Sub-tasks:	
1. Select time period	Admin chooses a specific timeframe (e.g. daily, monthly)
2. Retrieve sales data	System collects relevant sales information
3. Display sales summary	System shows total sales and trends
4. Identify best selling books	System highlights best selling books or categories
Variants:	

5. Workflow Diagrams

Task 1: Create Customer Account**Task 2: Sign in to System****Task 3: Search and Browse Books****Task 4: View Books Details**

Task 5: Add Books to Cart**Task 6: Modify Shopping Cart**

Task 7: Review and Confirm Order**Task 8: Complete Payment Process****Task 9: Generate Invoice and Receipt****Task 10: Check Order Process**

Task 11: Manage Book Listing and Classification**Task 12: Order Packaging and Delivery****Task 13: Analyse Sale Performance**

6. Product Design

6.1 Data / Domain Model

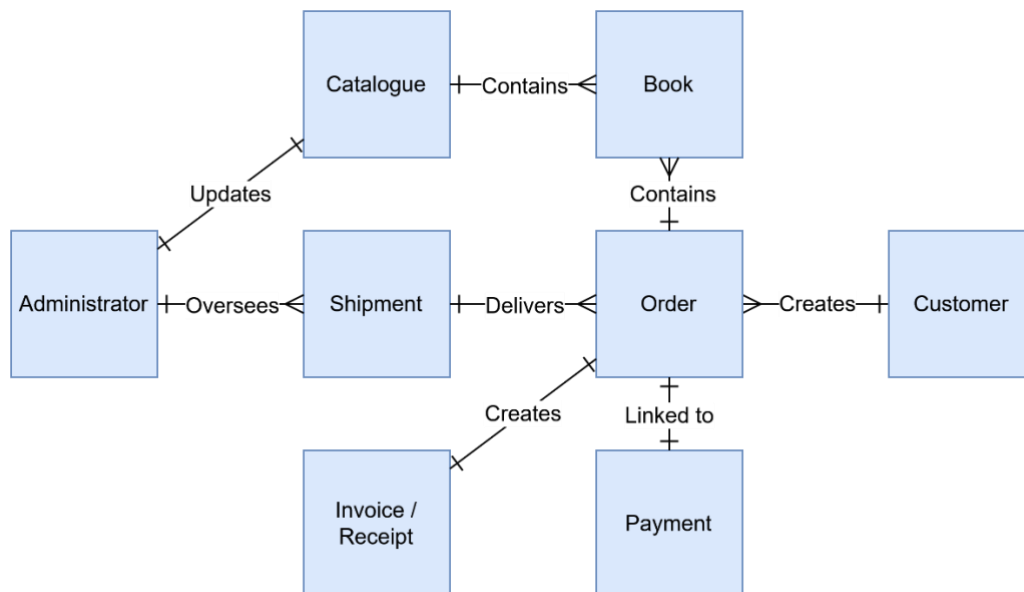


Figure 1 Domain Model of the System

6.2 Functional Requirements

The Favourite Books Online Store must provide the functions needed to support both customer-facing online purchasing and internal bookstore operations.

At the customer level, the system shall allow users to:

- create an account
- sign in securely
- browse and search the book catalogue
- view detailed book information
- add books to a shopping cart
- modify cart contents
- review and confirm orders
- complete payment
- receive an invoice and receipt
- check the progress of their orders

At the administrative level, the system shall allow bookstore staff to:

- manage book listings and classification
- update the catalogue as new books become available
- process packaging and shipment of completed orders
- analyse sales performance over selected time periods
- view sales statistics on a daily, weekly, monthly, yearly, and year-to-date basis

The system shall enforce basic operational integrity by:

- validating customer registration details
- ensuring that payment is confirmed before an order is treated as complete
- updating stock levels after a successful purchase
- maintaining accurate transaction records for invoices and receipts
- preventing checkout when required information is incomplete
- handling failed payment attempts by notifying the customer and allowing another attempt

6.3 Quality Attributes

Security

Security is critical because the system stores sensitive customer information such as account details, order history, and payment related to important data. If this information is compromised it may lead to financial loss, privacy breaches and damage to the business reputation.

Therefore, strong protection are required to ensure that only authorised users can access the system and that all transactions are handled securely.

To meet this requirement, the system shall:

- require secure authentication for both customers and administrative users
- restrict access to system functions based on user roles and permissions
- securely store sensitive data such as passwords using encryption
- prevent unauthorised access to administrative features such as catalogue management and reporting
- protect against common security risks such as invalid login attempts and unauthorised data access

Usability

Usability is important because the system is designed for general bookstore customers and staff with varying levels of technical experience. A system that is difficult to use may lead to user frustration, errors during checkout and reduced sales. The system must therefore provide a simple and clear interface that supports users in completing tasks efficiently.

To meet this requirement, the system shall:

- provide clear and navigation for browsing, searching and purchasing books
- allow users to easily manage their shopping cart and review orders before checkout
- display clear and helpful error messages for invalid inputs, failed payments or missing information
- support efficient completion of common tasks such as account registration, ordering and tracking deliveries
- follow existing user interface guidelines to ensure a familiar and standardised user experience

Performance

Performance is important because the system will be frequently used for browsing books, searching for items and completing purchases, so slow response times may negatively impact user experience and lead to abandoned transactions. The system must therefore provide fast and responsive performance under normal usage conditions.

To meet this requirement, the system shall:

- return search and browsing results within an acceptable response time under normal conditions
- process shopping cart updates and checkout operations without noticeable delay
- support multiple users accessing the system simultaneously without significant performance degradation
- generate administrative reports efficiently for selected time periods
- ensure that system response remains stable during peak usage periods

Reliability

Reliability is important because customers must be able to trust that orders, payments, invoices and delivery updates are recorded correctly. Any errors may lead to incorrect transactions, delays or customer dissatisfaction.

To meet this requirement, the system shall:

- preserve transaction accuracy for orders, payments and invoices
- prevent duplicate, incomplete or lost orders
- ensure that orders are only marked as completed after successful payment confirmation
- maintain accurate and up to date order status information
- update stock and shipment data consistently after each transaction
- handle failures such as payment errors without losing data

6.4 Other Requirements

Product Level Requirements

The Online Bookstore System shall be provided as a web-based software product that supports customer access through standard internet-connected devices and supports internal staff operations such as order processing and reporting.

To meet this requirement, the system shall:

- support core bookstore functions including catalogue browsing, ordering, payment handling, invoice and receipt generation, shipment support, and sales reporting
- limit the system scope to core bookstore operations only
- exclude features such as discounts, loyalty programs, and rent-before-buying in the current version
- allow future extensions without restricting the addition of new features
- integrate with an external payment service for secure transaction processing
- support integration with external delivery services for shipment handling
- operate as a software solution only, with hardware and deployment handled separately

Design Level Requirements

The system shall be designed in a modular structure so that key functions such as account management, browsing, shopping cart, order processing, payment, shipment and reporting are separated into distinct components.

To meet this requirement, the system shall:

- separate major system functions into independent modules to improve maintainability
- support future system expansion without requiring major redesign
- maintain consistency across customer and administrative interfaces
- enforce appropriate access control between customer and staff functions

- follow existing organisation interface guidelines for user interface design

7. Validation and Verifiability

Validation and verifiability are essential to ensure that the system meets user needs and that all requirements can be tested effectively. In this project, validation ensures that the Favourite Books Online Store is the correct solution by reviewing requirements against real-world bookstore operations, aligning them with stakeholder needs such as online sales, inventory management, and delivery tracking, and using scenario-based evaluation (e.g. browsing books, adding items to a cart, and completing payment) to confirm that all user workflows are fully supported without missing steps. Consistency checks are also applied to ensure there are no conflicts between requirements, such as maintaining correct stock updates after successful payments and accurate order tracking during delivery. Verifiability ensures that each requirement can be objectively tested through functional testing of system tasks (e.g. account creation, login, cart management, and payment processing), input and output validation using both valid and invalid data, and system behaviour testing under critical conditions such as payment failures or out-of-stock items. In addition, data integrity checks confirm that transactions, invoices, and stock levels are correctly recorded and updated, while performance verification ensures that key operations such as searching the catalogue and completing checkout occur within acceptable response times. Together, these validation and verification approaches ensure that the system is both correct in design and reliable in operation.

CRUD Check

Entity \ Task	Customer	Book	Catalogue	Order	Payment	Invoice/ Receipt	Shipment
Create Customer Account	C						
Sign in to System	R						
Search and Browse Books		R	R				
View Book Details		R					
Add Book to Cart		R					
Modify Shopping Cart		R					
Review and Confirm Order	R	R		C			
Complete Payment Process	R			R, U	C		

Generate Invoice and Receipt	R			R	R	C	
Check Order Process	R			R			R
Manage Book Listing and Classification		C, R, U, D	C, R, U, D				
Analyse Sales Performance				R	R	R	
Order Packaging and Delivery				R, U		R	C, U

8. Solutions

8.1 Possible Solution: Website-Based Online System

This solution aims to develop a website-based online bookstore system that would help the business expand its reach from a local store to across Australia. Currently, the majority of customers come from the surrounding area, with all transactions being done in person. Records are handled on paper without a proper managing system, which is suboptimal, reducing management efficiency. Therefore, by introducing a website, it would allow the business to extend its reach, improve customer experience through digitisation, and improve daily operations. The system would aim to improve accessibility with reliability, as customers can now place orders digitally. On top of this, the business would now have records of past orders and analytics to support management goals.

Once the system has been setup, the owners would be able to manage their inventory, update new arrival books, or update them to match their current stock level. This information is required to be up to date as incorrect display details could lead to failed orders and customer dissatisfaction. Furthermore, the website would allow customers to search, browse and view book details before ordering. This allows an easy and convenient purchasing process for customers.

To make a purchase, customers would be required to log in to their account, and new users would need to register beforehand. By doing this, order information and customers details can be recorded and securely stored in the system for analysis and tracking. Additional functions would also be provided such as adding books to cart, reviewing and editing shopping cart before proceeding to checkout. This additional stage would minimise errors and give customers the flexibility to review and adjust if needed. Following successful completion of the order through an external service, a confirmation of order would be presented to customer, and a copy of receipt would be sent to their email address. Due to the sensitive nature of online transactions, a secured handling is required to reduce risks such as unauthorised access or failed payments. This means, orders can only be processed after payment is successfully confirmed, and an invoice/receipt is generated and stored as a record of the transaction.

After successful payment has been confirmed, stock levels will be updated automatically to accurately reflect the available stock. Failure to update this could lead to overselling or delays in fulfilment of orders. Store staff could then proceed with packaging and preparing for shipment, with shipment details recorded in the system to track delivery progress. This would allow for transparency by allowing customers to sign into their accounts and track the progress of the order without having to raise an enquiry to the store directly.

Aside from supporting the customer shopping experience, the system would also support the business in managing their operations. The owners could login as administrators through the same platform to manage categories and update their stock records. This system aims to provides sales information over different time periods, which would provide the business the ability to review their performance and supports in decision making. This solution helps Favourite Books reach more customers, makes daily operations easier to manage, and supports the business as it continues to grow.

8.2 Possible Solution: Mobile app based solution

The development of a mobile based online bookstore system, which would be designed to offer a more accessible and customized user experience than a standard website, is a potential alternate solution. Customers would be able to easily browse books, search for specific titles, examine full details, and make purchases from their devices at any time with a mobile app thanks to the dominant popularity of smartphones. The app would offer key characteristics including account registration, safe login, shopping cart management, checkout, and payment processing through an outside payment service to guarantee transaction security, just as the online system.

Additionally, a mobile application could increase customer engagement and retention by providing improved features like push notifications to let users know about new arrivals, specials, or order status updates. Additionally, the system would keep data synchronized in real-time, guaranteeing that order details, transaction records, and stock levels would be regularly updated. On an administrative level, the backend system would be unchanged, enabling employees to effectively handle inventories, handle deliveries, and evaluate sales results. However, platform-specific factors would include compatibility with iOS and Android, and higher early development and maintenance expenses might be necessary when creating a mobile application. Despite these difficulties, the mobile app solution would offer better user interaction, increased convenience, and long-term scalability, making it a viable option for boosting client satisfaction and promoting business growth.